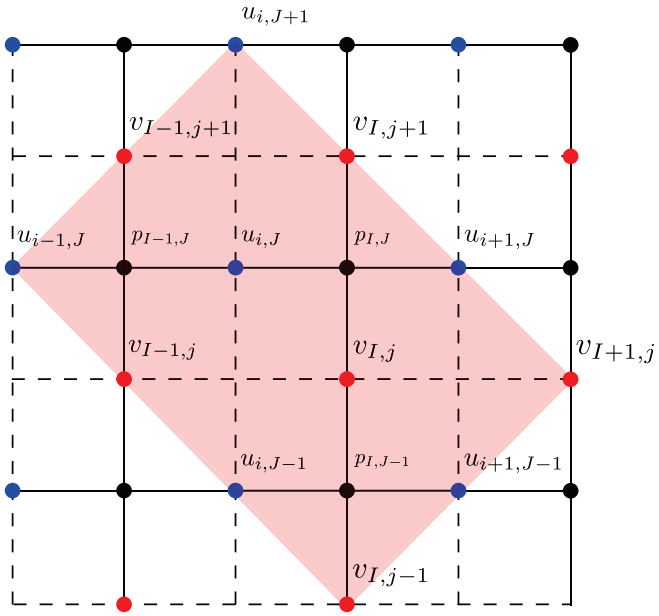


Fig. 1. One cell in the grid for which we define one x-momentum, one y-momentum and one continuity equation.

Linearised model of WFSim

I. Introduction

In this document we develop a linearised version of WFSim. The general approach is as follows. We write down the spatially discretised x-momentum, y-momentum and continuity equation for one cell. A cell is defined as in Fig. 1. Then we define a state variable for the three equations containing all the state variables which occur in the three equations. Then, a standard linearisation technique is applied and at last, we lift the state vector such that we get a linearised version of the complete field.

II. General Formulation

According to Fig. 1 we can write down the following equations:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}u_{i,J} &= a_x(x) + a_{xc}(p_{I-1,J} - p_{I,J}) + S^x_{i,J}, && \text{x-momentum}\\
\frac{\mathrm{d}}{\mathrm{d}t}v_{I,j} &= a_y(x) + a_{yc}(p_{I,J-1} - p_{I,J}) + S^y_{I,j}, && \text{y-momentum}\\
0 &= c_1\left(u_{i+1,J} - u_{i,J}\right) + c_2\left(v_{I,j+1} - v_{I,j}\right), && \text{continuity}
\end{aligned}$$

with

$$\begin{aligned}
a_x(x) &= -(a^{P_x} + T^{N_x} + T^{S_x})u_{i,J} + a^{W_x}u_{i-1,J} + a^{E_x}u_{i+1,J} + (a^{N_x} + T^{N_x})u_{i,J+1} + (a^{S_x} + T^{S_x})u_{i,J-1}\\
a_y(x) &= -a^{P_y}v_{I,j} + a^{W_y}v_{I-1,j} + a^{E_y}v_{I+1,j} + a^{N_y}v_{I,j+1} + a^{S_y}v_{I,j-1}
\end{aligned}$$

and a_{xc}, a_{yc}, c_1, c_2 constant coefficients depending on the grid. Furthermore we have that all a coefficients depend on certain velocity components. We have

$$\begin{aligned}
&a^{P_x}(u_{i-1,J}, u_{i,J}, u_{i+1,J}, v_{I-1,j}, v_{I-1,j+1}, v_{I,j}, v_{I,j+1}),\ a^{W_x}(u_{i-1,J}, u_{i,J}),\ a^{E_x}(u_{i,J}, u_{i+1,J}),\ a^{N_x}(v_{I-1,j+1}, v_{I,j+1}),\ a^{S_x}(v_{I-1,j}, v_{I,j})\\
&a^{P_y}(u_{i,J-1}, u_{i,J}, u_{i+1,J-1}, u_{i+1,J}, v_{I,j-1}, v_{I,j}, v_{I,j+1}),\ a^{W_y}(u_{i,J-1}, u_{i,J}),\ a^{E_y}(u_{i+1,J-1}, u_{i+1,J}),\ a^{N_y}(v_{I,j+1}, v_{I,j}),\ a^{S_y}(v_{I,j}, v_{I,j-1})
\end{aligned}$$

The turbulence terms are defined as:

$$T^{N_x} = c_n\left|u_{i,J+1} - u_{i,J}\right|, \qquad T^{S_x} = c_s\left|u_{i,J} - u_{i,J-1}\right| \tag{1}$$

Accordingly we can define a reduced state x_r, a state x and input w as

$$x_r = \begin{pmatrix} u_{i,J} & v_{I,j} \end{pmatrix}^T$$

$$x = \left(\begin{array}{cccccc|cccccc|ccc} u_{i-1,J} & u_{i,J-1} & u_{i,J} & u_{i,J+1} & u_{i+1,J-1} & u_{i+1,J} & v_{I-1,j} & v_{I-1,j+1} & v_{I,j-1} & v_{I,j} & v_{I,j+1} & v_{I+1,j+1} & p_{I-1,J} & p_{I,J-1} & p_{I,J} \end{array} \right)^T$$

$$w = \begin{pmatrix} S^x_{i,J} & S^y_{I,j} \end{pmatrix}^T$$

The general formulation for one cell is

$$\begin{aligned} \dot{x}_r &= f(x) + g(w) \\ 0 &= h(x) \end{aligned} \tag{2}$$

with

$$f(x) = \begin{pmatrix} a_x(x) + a_{xc}(p_{I-1,J} - p_{I,J}) \\ a_y(x) + a_{yc}(p_{I,J-1} - p_{I,J}) \end{pmatrix}, \quad g(w) = \begin{pmatrix} S^x_{i,J} \\ S^y_{I,j} \end{pmatrix}, \quad h(x) = c_1 \left(u_{i+1,J} - u_{i,J}\right) + c_2 \left(v_{I,j+1} - v_{I,j}\right)$$
$$= \begin{pmatrix} f_1(x) \\ f_2(x) \end{pmatrix}$$

If we linearise we get the following linearised model for one cell:

$$\dot{x}_r = \underbrace{\left.\frac{\partial f(x)}{\partial x}\right|_{x_0}}_{A} \delta x + \underbrace{\left.\frac{\partial g(w)}{\partial w}\right|_{w_0}}_{B} \delta w, \qquad 0 = h(\delta x) \tag{3}$$

with $\delta x, \delta w$ the deviation variables and x_0 and w_0 the points of linearisation. We have as deviation state variable:

$$\delta x = \left(\begin{array}{cccccc|cccccc|c} \delta u_{i-1,J} & \delta u_{i,J-1} & \delta u_{i,J} & \delta u_{i,J+1} & \delta u_{i+1,J-1} & \delta u_{i+1,J} & \delta v_{I-1,j} & \delta v_{I-1,j+1} & \delta v_{I,j-1} & \delta v_{I,j} & \delta v_{I,j+1} & \delta v_{I+1,j} & \dots \\ \dots & \delta p_{I-1,J} & \delta p_{I,J-1} & \delta p_{I,J} & & & & & & & & & \end{array} \right)$$

Furthermore we have

$$\frac{\partial f(x)}{\partial x} = \begin{pmatrix} f_1(x) \\ f_2(x) \end{pmatrix} \left(\begin{array}{cccccc|cccccc|c} \frac{\partial}{\partial u_{i-1,J}} & \frac{\partial}{\partial u_{i,J-1}} & \frac{\partial}{\partial u_{i,J}} & \frac{\partial}{\partial u_{i,J+1}} & \frac{\partial}{\partial u_{i+1,J-1}} & \frac{\partial}{\partial u_{i+1,J}} & \frac{\partial}{\partial v_{I-1,j}} & \frac{\partial}{\partial v_{I-1,j+1}} & \frac{\partial}{\partial v_{I,j-1}} & \frac{\partial}{\partial v_{I,j}} & \frac{\partial}{\partial v_{I,j+1}} & \frac{\partial}{\partial v_{I+1,j+1}} & \dots \\ \dots & \frac{\partial}{\partial p_{I-1,J}} & \frac{\partial}{\partial p_{I,J-1}} & \frac{\partial}{\partial p_{I,J}} & & & & & & & & & \end{array} \right)$$

$$\frac{\partial g(w)}{\partial w} = \begin{pmatrix} g_1(w) \\ g_2(w) \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial w_1} & \frac{\partial}{\partial w_2} \end{pmatrix}$$

In Eq. 3 we have a linearised model for one cell. If we lift the state variable to a deviation state variable $\delta\tilde{x}$ we can obtain a set of equations containing the linearised models of all the cells in the field.

In the following subsections we will compute the partial derivatives of first the x-momentum and then the y-momentum equation.

A. *x-momentum*

We have the spatially discretised x-momentum equation

$$\frac{\mathrm{d}}{\mathrm{d}t}u_{i,J} = \underbrace{a_x(x) + a_{xc}(p_{I-1,J} - p_{I,J})}_{f_1(x)} + S^x_{i,J} \tag{4}$$

with

$$a_x(x) = -(a^{P_x} + T^{N_x} + T^{S_x})u_{i,J} + a^{W_x}u_{i-1,J} + a^{E_x}u_{i+1,J} + (a^{N_x} + T^{N_x})u_{i,J+1} + (a^{S_x} + T^{S_x})u_{i,J-1} \tag{5}$$

We have the following dependency of the coefficients:
$a^{P_x}(u_{i-1,J}, u_{i,J}, u_{i+1,J},\ v_{I-1,j}, v_{I-1,j+1}, v_{I,j}, v_{I,j+1})$, $a^{W_x}(u_{i-1,J}, u_{i,J})$, $a^{E_x}(u_{i,J}, u_{i+1,J})$, $a^{N_x}(v_{I-1,j+1}, v_{I,j+1})$, $a^{S_x}(v_{I-1,j}, v_{I,j})$
$T^{N_x}(u_{i,J+1}, u_{i,J})$, $T^{S_x}(u_{i,J}, u_{i,J-1})$

When using Eq. 3 we get:

$$\frac{\partial f_1(x)}{\partial x} = \Bigg(\frac{\partial f_1(x)}{\partial u_{i-1,J}} \quad \frac{\partial f_1(x)}{\partial u_{i,J-1}} \quad \frac{\partial f_1(x)}{\partial u_{i,J}} \quad \frac{\partial f_1(x)}{\partial u_{i,J+1}} \quad \frac{\partial f_1(x)}{\partial u_{i+1,J-1}} \quad \frac{\partial f_1(x)}{\partial u_{i+1,J}} \;\Big|\; \frac{\partial f_1(x)}{\partial v_{I-1,j}} \quad \frac{\partial f_1(x)}{\partial v_{I-1,j+1}} \quad \frac{\partial f_1(x)}{\partial v_{I,j-1}} \quad \frac{\partial f_1(x)}{\partial v_{I,j}} \quad \frac{\partial f_1(x)}{\partial v_{I,j+1}} \quad \frac{\partial f_1(x)}{\partial v_{I+1,j}} \;\Big|\; \dots$$
$$\dots \quad \frac{\partial f_1(x)}{\partial p_{I-1,J}} \quad \frac{\partial f_1(x)}{\partial p_{I,J-1}} \quad \frac{\partial f_1(x)}{\partial p_{I,J}} \Bigg)$$

with

$$\begin{aligned}
\frac{\partial f_1(x)}{\partial u_{i-1,J}} &= -\frac{\partial a^{P_x}}{\partial u_{i-1,J}}u_{i,J} + a^{W_x} + \frac{\partial a^{W_x}}{\partial u_{i-1,J}}u_{i-1,J}, \ (= \delta a^{wx}) \\
\frac{\partial f_1(x)}{\partial u_{i,J-1}} &= -\frac{\partial T^{S_x}}{\partial u_{i,J-1}}u_{i,J} + a^{S_x} + \frac{\partial T^{S_x}}{\partial u_{i,J-1}}u_{i,J-1} + T^{S_x}, \ (= \delta a^{sx}) \\
\frac{\partial f_1(x)}{\partial u_{i,J}} &= -\frac{\partial a^{P_x}}{\partial u_{i,J}}u_{i,J} - a^{P_x} + \frac{\partial a^{W_x}}{\partial u_{i,J}}u_{i-1,J} + \frac{\partial a^{E_x}}{\partial u_{i,J}}u_{i+1,J} + \\
&\quad -\frac{\partial T^{N_x}}{\partial u_{i,J}}u_{i,J} - T^{N_x} - \frac{\partial T^{S_x}}{\partial u_{i,J}}u_{i,J} - T^{S_x} + \frac{\partial T^{N_x}}{\partial u_{i,J}}u_{i,J+1} + \frac{\partial T^{S_x}}{\partial u_{i,J}}u_{i,J-1} \ (= \delta a^{px}) \\
\frac{\partial f_1(x)}{\partial u_{i,J+1}} &= -\frac{\partial T^{N_x}}{\partial u_{i,J+1}}u_{i,J} + a^{N_x} + \frac{\partial T^{N_x}}{\partial u_{i,J+1}}u_{i,J+1} + T^{N_x}, \ (= \delta a^{nx}) \\
\frac{\partial f_1(x)}{\partial u_{i+1,J-1}} &= 0, \\
\frac{\partial f_1(x)}{\partial u_{i+1,J}} &= -\frac{\partial a^{P_x}}{\partial u_{i+1,J}}u_{i,J} + \frac{\partial a^{E_x}}{\partial u_{i+1,J}}u_{i+1,J} + a^{E_x}, \ (= \delta a^{ex}) \\
\frac{\partial f_1(x)}{\partial v_{I-1,j}} &= -\frac{\partial a^{P_x}}{\partial v_{I-1,j}}u_{i,J} + \frac{\partial a^{S_x}}{\partial v_{I-1,j}}u_{i,J-1}, \ (= \delta a^{swx}) \\
\frac{\partial f_1(x)}{\partial v_{I-1,j+1}} &= -\frac{\partial a^{P_x}}{\partial v_{I-1,j+1}}u_{i,J} + \frac{\partial a^{N_x}}{\partial v_{I-1,j+1}}u_{i,J+1}, \ (= \delta a^{nwx}) \\
\frac{\partial f_1(x)}{\partial v_{I,j-1}} &= 0, \\
\frac{\partial f_1(x)}{\partial v_{I,j}} &= -\frac{\partial a^{P_x}}{\partial v_{I,j}}u_{i,J} + \frac{\partial a^{S_x}}{\partial v_{I,j}}u_{i,J-1}, \ (= \delta a^{sex}) \\
\frac{\partial f_1(x)}{\partial v_{I,j+1}} &= -\frac{\partial a^{P_x}}{\partial v_{I,j+1}}u_{i,J} + \frac{\partial a^{N_x}}{\partial v_{I,j+1}}u_{i,J+1}, \ (= \delta a^{nex}) \\
\frac{\partial f_1(x)}{\partial v_{I+1,j}} &= 0,
\end{aligned}$$

$$\frac{\partial f_1(x)}{\partial p_{I-1,J}} = a_{xc},$$
$$\frac{\partial f_1(x)}{\partial p_{I,J-1}} = 0,$$
$$\frac{\partial f_1(x)}{\partial p_{I,J}} = -a_{xc},$$

Note that all the partials are evaluated at an equilibrium.

B. y-momentum

We have the spatially discretised y-momentum equation

$$\frac{\mathrm{d}}{\mathrm{d}t}v_{I,j} = \underbrace{a_y(x) + a_{yc}(p_{I,J-1} - p_{I,J})}_{f_2(x)} + S^y_{I,j} \tag{6}$$

with

$$a_y(x) = -a^{P_y}v_{I,j} + a^{W_y}v_{I-1,j} + a^{E_y}v_{I+1,j} + a^{N_y}v_{I,j+1} + a^{S_y}v_{I,j-1} \tag{7}$$

We have the following dependency of the coefficients:
$a^{P_y}(u_{i,J-1}, u_{i,J}, u_{i+1,J-1}, u_{i+1,J}, v_{I,j-1}, v_{I,j}, v_{I,j+1})$, $a^{W_y}(u_{i,J-1}, u_{i,J})$, $a^{E_y}(u_{i+1,J-1}, u_{i+1,J})$, $a^{N_y}(v_{I,j-1}, v_{I,j})$, $a^{S_y}(v_{I,j}, v_{I,j+1})$.

When using Eq. 3 we get:

$$\frac{\partial f_2(x)}{\partial x} = \left(\begin{array}{cccccc|cccccc|c} \frac{\partial f_2(x)}{\partial u_{i-1,J}} & \frac{\partial f_2(x)}{\partial u_{i,J-1}} & \frac{\partial f_2(x)}{\partial u_{i,J}} & \frac{\partial f_2(x)}{\partial u_{i,J+1}} & \frac{\partial f_2(x)}{\partial u_{i+1,J-1}} & \frac{\partial f_2(x)}{\partial u_{i+1,J}} & \frac{\partial f_2(x)}{\partial v_{I-1,j}} & \frac{\partial f_2(x)}{\partial v_{I-1,j+1}} & \frac{\partial f_2(x)}{\partial v_{I,j-1}} & \frac{\partial f_2(x)}{\partial v_{I,j}} & \frac{\partial f_2(x)}{\partial v_{I,j+1}} & \frac{\partial f_2(x)}{\partial v_{I+1,j}} & \cdots \\ \cdots & \frac{\partial f_2(x)}{\partial p_{I-1,J}} & \frac{\partial f_2(x)}{\partial p_{I,J-1}} & \frac{\partial f_2(x)}{\partial p_{I,J}} & & & & & & & & & & \end{array} \right)$$

with

$$\begin{aligned}
\frac{\partial f_2(x)}{\partial u_{i-1,J}} &= 0, \\
\frac{\partial f_2(x)}{\partial u_{i,J-1}} &= -\frac{\partial a^{P_y}}{\partial u_{i,J-1}}v_{I,j} + \frac{\partial a^{W_y}}{\partial u_{i,J-1}}v_{I-1,j}, \ (= \delta a^{swy}) \\
\frac{\partial f_2(x)}{\partial u_{i,J}} &= -\frac{\partial a^{P_y}}{\partial u_{i,J}}v_{I,j} + \frac{\partial a^{W_y}}{\partial u_{i,J}}v_{I-1,j}, \ (= \delta a^{nwy}) \\
\frac{\partial f_2(x)}{\partial u_{i,J+1}} &= 0, \\
\frac{\partial f_2(x)}{\partial u_{i+1,J-1}} &= -\frac{\partial a^{P_y}}{\partial u_{i+1,J-1}}v_{I,j} + \frac{\partial a^{E_y}}{\partial u_{i+1,J-1}}v_{I+1,j}, \ (= \delta a^{sey}) \\
\frac{\partial f_2(x)}{\partial u_{i+1,J}} &= -\frac{\partial a^{P_y}}{\partial u_{i+1,J}}v_{I,j} + \frac{\partial a^{E_y}}{\partial u_{i+1,J}}v_{I+1,j}, \ (= \delta a^{ney}) \\
\frac{\partial f_2(x)}{\partial v_{I-1,j}} &= a^{W_y}, \ (= \delta a^{wy}) \\
\frac{\partial f_2(x)}{\partial v_{I-1,j+1}} &= 0, \\
\frac{\partial f_2(x)}{\partial v_{I,j-1}} &= -\frac{\partial a^{P_y}}{\partial v_{I,j-1}}v_{I,j} + \frac{\partial a^{S_y}}{\partial v_{I,j-1}}v_{I,j-1} + a^{S_y}, \ (= \delta a^{sy}) \\
\frac{\partial f_2(x)}{\partial v_{I,j}} &= -\frac{\partial a^{P_y}}{\partial v_{I,j}}v_{I,j} - a^{P_y} + \frac{\partial a^{N_y}}{\partial v_{I,j}}v_{I,j+1} + \frac{\partial a^{S_y}}{\partial v_{I,j}}v_{I,j-1}, \ (= \delta a^{py}) \\
\frac{\partial f_2(x)}{\partial v_{I,j+1}} &= -\frac{\partial a^{P_y}}{\partial v_{I,j+1}}v_{I,j} + \frac{\partial a^{N_y}}{\partial v_{I,j+1}}v_{I,j+1} + a^{N_y}, \ (= \delta a^{ny}) \\
\frac{\partial f_2(x)}{\partial v_{I+1,j}} &= a^{E_y}, \ (= \delta a^{ey}), \\
\frac{\partial f_2(x)}{\partial p_{I-1,J}} &= 0, \\
\frac{\partial f_2(x)}{\partial p_{I,J-1}} &= a_{yc}, \\
\frac{\partial f_2(x)}{\partial p_{I,J}} &= -a_{yc},
\end{aligned}$$

Note that all the partials are evaluated at an equilibrium.

C. Continuity equation

The linearised continuity equation is defined as:

$$\begin{aligned} 0 &= c_1 \left(\delta u_{i+1,J} - \delta u_{i,J}\right) + c_2 \left(\delta v_{I,j+1} - \delta v_{I,j}\right), \\ &= h(\delta x) \end{aligned}$$

III. Temporal discretization and lifting the state vector

Finally we can obtain, after temporal discretisation and lifting the state vector to $\delta\tilde{x}$, the following linearised equation for the complete field:

$$\begin{pmatrix} A_1 & A_2 & B_1 \\ A_3 & A_4 & B_2 \\ B_1^T & B_2^T & 0 \end{pmatrix} \delta\tilde{x}_{k+1} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \tag{8}$$

IV. Linearised qLPV model

In this section we will transform the system given in Eq. 8 to a qLPV model. This will result in

$$E\,\alpha_{k+1} = A\alpha_k + B\beta_k + S \tag{9}$$

$$\delta\tilde{x}_k = Q_p\alpha_k + B_p \tag{10}$$

In order to get a linearised qLPV model we let WFSim find a steady state solution and use this steady state solution to construct the system matrices. Then, a time simulation can be performed using the qLPV model.

V. Example

$$
\left(\begin{array}{cccc|cccc|cccc|cccc|cccc|cccc}
-\delta a^{px}_{3,2} & \delta a^{nx}_{3,2} & 0 & 0 & \delta a^{ex}_{3,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{nwx}_{3,2} & 0 & 0 & \delta a^{nex}_{3,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\delta a^{sx}_{3,3} & -\delta a^{px}_{3,3} & \delta a^{nx}_{3,3} & 0 & 0 & \delta a^{ex}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{3,3} & \delta a^{nwx}_{3,3} & 0 & \delta a^{sex}_{3,3} & \delta a^{nex}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \delta a^{sx}_{3,4} & -\delta a^{px}_{3,4} & \delta a^{nx}_{3,4} & 0 & 0 & \delta a^{ex}_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{3,4} & \delta a^{nwx}_{3,4} & 0 & \delta a^{sex}_{3,4} & \delta a^{nex}_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \delta a^{sx}_{3,5} & -\delta a^{px}_{3,5} & 0 & 0 & 0 & \delta a^{ex}_{3,5} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{3,5} & 0 & 0 & \delta a^{sex}_{3,5} & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
\delta a^{wx}_{4,2} & 0 & 0 & 0 & -\delta a^{px}_{4,2} & \delta a^{nx}_{4,2} & 0 & 0 & \delta a^{ex}_{4,2} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{nwx}_{4,2} & 0 & 0 & \delta a^{nex}_{4,2} & 0 & 0 & 0 & 0 & 0 \\
0 & \delta a^{wx}_{4,3} & 0 & 0 & \delta a^{sx}_{4,3} & -\delta a^{px}_{4,3} & \delta a^{nx}_{4,3} & 0 & 0 & \delta a^{ex}_{4,3} & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{4,3} & \delta a^{nwx}_{4,3} & 0 & \delta a^{sex}_{4,3} & \delta a^{nex}_{4,3} & 0 & 0 & 0 & 0 \\
0 & 0 & \delta a^{wx}_{4,4} & 0 & 0 & \delta a^{sx}_{4,4} & -\delta a^{px}_{4,4} & \delta a^{nx}_{4,4} & 0 & 0 & \delta a^{ex}_{4,4} & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{4,4} & \delta a^{nwx}_{4,4} & 0 & \delta a^{sex}_{4,4} & \delta a^{nex}_{4,4} & 0 & 0 & 0 \\
0 & 0 & 0 & \delta a^{wx}_{4,5} & 0 & 0 & \delta a^{sx}_{4,5} & -\delta a^{px}_{4,5} & 0 & 0 & 0 & \delta a^{ex}_{4,5} & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{4,5} & 0 & 0 & \delta a^{sex}_{4,5} & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & \delta a^{wx}_{5,2} & 0 & 0 & 0 & -\delta a^{px}_{5,2} & \delta a^{nx}_{5,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{nwx}_{5,2} & 0 & 0 & \delta a^{nex}_{5,2} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \delta a^{wx}_{5,3} & 0 & 0 & \delta a^{sx}_{5,3} & -\delta a^{px}_{5,3} & \delta a^{nx}_{5,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{5,3} & \delta a^{nwx}_{5,3} & 0 & \delta a^{sex}_{5,3} & \delta a^{nex}_{5,3} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wx}_{5,4} & 0 & 0 & \delta a^{sx}_{5,4} & -\delta a^{px}_{5,4} & \delta a^{nx}_{5,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{5,4} & \delta a^{nwx}_{5,4} & 0 & \delta a^{sex}_{5,4} & \delta a^{nex}_{5,4} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wx}_{5,5} & 0 & 0 & \delta a^{sx}_{5,5} & -\delta a^{px}_{5,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{5,5} & 0 & 0 & \delta a^{sex}_{5,5} \\
\hline
\delta a^{sey}_{2,3} & \delta a^{ney}_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta a^{py}_{2,3} & \delta a^{ny}_{2,3} & 0 & \delta a^{ey}_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \delta a^{sey}_{2,4} & \delta a^{ney}_{2,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{sy}_{2,4} & -\delta a^{py}_{2,4} & \delta a^{ny}_{2,4} & 0 & \delta a^{ey}_{2,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \delta a^{sey}_{2,5} & \delta a^{ney}_{2,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{sy}_{2,5} & -\delta a^{py}_{2,5} & 0 & 0 & \delta a^{ey}_{2,5} & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
\delta a^{swy}_{3,3} & \delta a^{nwy}_{3,3} & 0 & 0 & \delta a^{sey}_{3,3} & \delta a^{ney}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{3,3} & 0 & 0 & -\delta a^{py}_{3,3} & \delta a^{ny}_{3,3} & 0 & \delta a^{ey}_{3,3} & 0 & 0 & 0 & 0 & 0 \\
0 & \delta a^{swy}_{3,4} & \delta a^{nwy}_{3,4} & 0 & 0 & \delta a^{sey}_{3,4} & \delta a^{ney}_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{3,4} & 0 & \delta a^{sy}_{3,4} & -\delta a^{py}_{3,4} & \delta a^{ny}_{3,4} & 0 & \delta a^{ey}_{3,4} & 0 & 0 & 0 & 0 \\
0 & 0 & \delta a^{swy}_{3,5} & \delta a^{nwy}_{3,5} & 0 & 0 & \delta a^{sey}_{3,5} & \delta a^{ney}_{3,5} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{3,5} & 0 & \delta a^{ny}_{3,5} & -\delta a^{py}_{3,5} & 0 & 0 & \delta a^{ey}_{3,5} & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & \delta a^{swy}_{4,3} & \delta a^{nwy}_{4,3} & 0 & 0 & \delta a^{sey}_{4,3} & \delta a^{ney}_{4,3} & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{4,3} & 0 & 0 & -\delta a^{py}_{4,3} & \delta a^{ny}_{4,3} & 0 & \delta a^{ey}_{4,3} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \delta a^{swy}_{4,4} & \delta a^{nwy}_{4,4} & 0 & 0 & \delta a^{sey}_{4,4} & \delta a^{ney}_{4,4} & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{4,4} & 0 & \delta a^{sy}_{4,4} & -\delta a^{py}_{4,4} & \delta a^{ny}_{4,4} & 0 & \delta a^{ey}_{4,4} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swy}_{4,5} & \delta a^{nwy}_{4,5} & 0 & 0 & \delta a^{sey}_{4,5} & \delta a^{ney}_{4,5} & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{4,5} & 0 & \delta a^{sy}_{4,5} & -\delta a^{py}_{4,5} & 0 & 0 & \delta a^{ey}_{4,5} \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swy}_{5,3} & \delta a^{nwy}_{5,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{5,3} & 0 & 0 & -\delta a^{py}_{5,3} & \delta a^{ny}_{5,3} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swy}_{5,4} & \delta a^{nwy}_{5,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{5,4} & 0 & \delta a^{sy}_{5,4} & -\delta a^{py}_{5,4} & \delta a^{ny}_{5,4} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swy}_{5,5} & \delta a^{nwy}_{5,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{5,5} & 0 & \delta a^{sy}_{5,5} & -\delta a^{py}_{5,5}
\end{array}\right)
\begin{bmatrix}
\delta u_{3,2} \\ \delta u_{3,3} \\ \delta u_{3,4} \\ \delta u_{3,5} \\ \hline
\delta u_{4,2} \\ \delta u_{4,3} \\ \delta u_{4,4} \\ \delta u_{4,5} \\ \hline
\delta u_{5,2} \\ \delta u_{5,3} \\ \delta u_{5,4} \\ \delta u_{5,5} \\ \hline
\delta v_{2,3} \\ \delta v_{2,4} \\ \delta v_{2,5} \\ \hline
\delta v_{3,3} \\ \delta v_{3,4} \\ \delta v_{3,5} \\ \hline
\delta v_{4,3} \\ \delta v_{4,4} \\ \delta v_{4,5} \\ \hline
\delta v_{5,3} \\ \delta v_{5,4} \\ \delta v_{5,5}
\end{bmatrix}
+\ldots \tag{11}
$$

$$
\cdots + \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} \delta\tilde{p} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \tag{12}
$$

and

$$
B_1^T \delta\tilde{u} + B_2^T \delta\tilde{v} = b_3 \tag{13}
$$

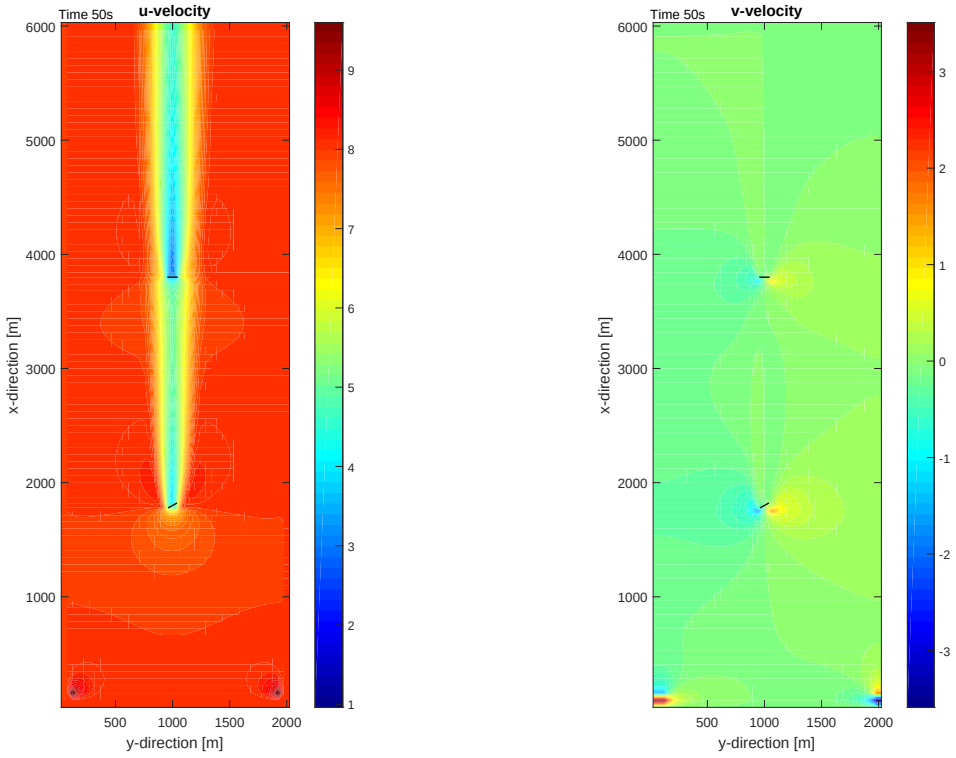


Fig. 2. Flow field generated after simulating the linear model.

VI. Results

In this section we present results obtained with the linearised model.

A. Flow field

In this section we present a flow field obtained using the linearised model. We let the non linear model converge to a steady state. This steady state field is used to construct the state matrices of the linear model. We then let the yaw angle of the first turbine go from 0 to 40 degrees and simulate for 50 seconds. So in the time loop, only the dynamical term and the actuator are updated. The final result is depicted in Fig. 2.

The following observations:

- Strange solutions in the corner for low x value.
- It seems that the force on the flow field due to the turbines is not high enough. In the non linear case we have $F = cu_r^2$ while in the linear case we have something like $F = cu_r$.
- If the solutions is better, a comparison can be made between the linear and non linear model.
- It seems that the projection can still be used in the linearised case. Both give equivalent results.
- Check if the state matrices make sense (e.g. look at Bode).

B. Bode magnitude plot

In this section we will transform the linearised WFSim to a qLPV model using a transformation. This result in the following linearised system.

$$E\alpha_{k+1} = A\alpha_k + B\beta_k + S \tag{14}$$

$$\delta\tilde{x}_k = Q_p\alpha_k + B_p \tag{15}$$

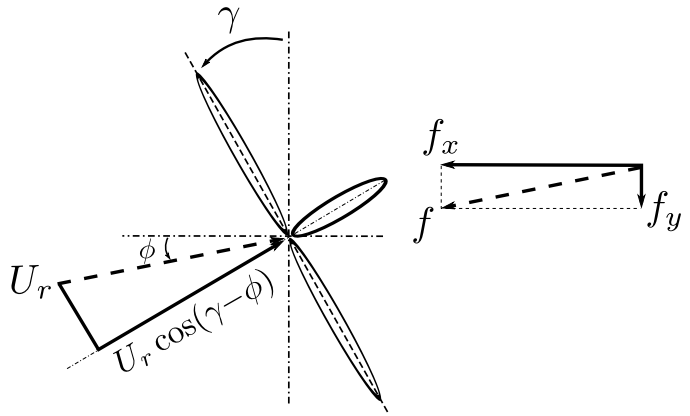


Fig. 3. Turbine seen from above in yawed position γ and U_r being the velocity and ϕ the wind direction for one cell at the rotor.

VII. Derivative of S_m to control inputs

A schematic representation of a yawed turbine is depicted in Fig 3. Here we have γ the yaw angle relative to the y-axis and ϕ the angle of the flow relative to the x-axis. We can define the force for one cell at the rotor now as:

$$S_m = C_T \frac{1}{2}\rho A U_{\text{effect}}^2 \tag{16}$$

Next we define the following relations:

$$\beta = \frac{a}{1-a}, \qquad U_{\text{effect}} = \frac{U_r}{1-a}, \qquad C_T = 4a(1-a)\cos(\gamma-\phi), \qquad U_r = \sqrt{u_r^2+v_r^2} \tag{17}$$

$$\tag{18}$$

Substituting these relations in (16) yields the following expression for the force S_m:

$$S_m = 2\rho A\beta U_r^2 \cos(\gamma-\phi) \tag{19}$$

Now we can define the force in the x- and y-direction as

$$S_m^x = -f\cos(\gamma), \qquad S_m^y = f\sin(\gamma) \tag{20}$$

$$\tag{21}$$

Now we define the following derivatives:

$$\frac{\partial S_m^x}{\partial \beta} = -2\rho A U_r^2 \cos(\gamma-\phi)\cos(\gamma) \tag{22}$$

$$\frac{\partial S_m^y}{\partial \beta} = 2\rho A U_r^2 \cos(\gamma-\phi)\sin(\gamma) \tag{23}$$

$$\frac{\partial S_m^x}{\partial \gamma} = 2\rho A\beta U_r^2 \Big(\sin(\gamma-\phi)\cos(\gamma)+\cos(\gamma-\phi)\sin(\gamma)\Big) \tag{24}$$

$$\frac{\partial S_m^y}{\partial \gamma} = 2\rho A\beta U_r^2 \Big(\cos(\gamma-\phi)\cos(\gamma)-\sin(\gamma-\phi)\sin(\gamma)\Big) \tag{25}$$

$$\frac{\partial S_m^x}{\partial x_k} = -4\rho A\beta (u_r+v_r)\cos(\gamma-\phi)\cos(\gamma) \tag{26}$$

$$\frac{\partial S_m^y}{\partial x_k} = 4\rho A\beta (u_r+v_r)\cos(\gamma-\phi)\sin(\gamma) \tag{27}$$

$$\tag{28}$$

VIII. Linearised model

We have the following set of equations:

$$A(x_k)x_{k+1} = Qx_k + B_c(x_k) + S_m(x_k, \beta, \gamma) \tag{29}$$

with

$$x = \begin{pmatrix} u \\ v \\ p \end{pmatrix}, \quad u = \begin{pmatrix} u_{3,2} \\ u_{3,3} \\ \vdots \\ u_{3,N_y-1} \\ u_{4,2} \\ \vdots \\ u_{N_x-1,N_y-1} \end{pmatrix}, \quad v = \begin{pmatrix} v_{2,3} \\ v_{2,4} \\ \vdots \\ v_{2,N_y-1} \\ v_{3,3} \\ \vdots \\ v_{N_x-1,N_y-1} \end{pmatrix}, \quad p = \begin{pmatrix} p_{2,2} \\ p_{2,3} \\ \vdots \\ p_{2,N_y-1} \\ p_{3,2} \\ \vdots \\ p_{N_x-1,N_y-1} \end{pmatrix} \tag{30}$$

and

$$A(x_k) = \begin{bmatrix} A_x(x_k) & 0 & B_1 \\ 0 & A_y(x_k) & B_2 \\ B_1^T & B_2^T & 0 \end{bmatrix}, \quad B_c(x_k) = \begin{pmatrix} a^{W_x}_{2,2}(u_b, u_{3,2})u_b \\ a^{W_x}_{2,3}(u_b, u_{3,3})u_b \\ \vdots \\ a^{W_x}_{2,N_y-1}(u_b, u_{3,N_y-1})u_b \\ 0 \\ \vdots \\ 0 \\ a^{W_y}_{1,3}(u_{3,1}, u_{3,2})v_b \\ a^{W_y}_{1,4}(u_{3,2}, u_{3,3})v_b \\ \vdots \\ a^{W_y}_{1,N_y-1}(u_{3,N_y-2}, u_{3,N_y-1})v_b \\ 0 \\ \vdots \\ 0 \\ c_1 u_b \\ c_2 u_b \\ \vdots \\ c_{N_y-1}u_b \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad S_m(x_k, \beta, \gamma) = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ -f(u_{r_1}, v_{r_1}, \beta_1)\cos(\gamma_1) \\ -f(u_{r_2}, v_{r_2}, \beta_1)\cos(\gamma_1) \\ 0 \\ 0 \\ \vdots \\ 0 \\ -f(u_{r_3}, v_{r_3}, \beta_2)\cos(\gamma_2) \\ -f(u_{r_4}, v_{r_4}, \beta_2)\cos(\gamma_2) \\ 0 \\ \vdots \\ 0 \\ f(u_{r_1}, v_{r_1}, \beta_1)\sin(\gamma_1) \\ f(u_{r_2}, v_{r_2}, \beta_1)\sin(\gamma_1) \\ 0 \\ 0 \\ \vdots \\ 0 \\ f(u_{r_3}, v_{r_3}, \beta_2)\sin(\gamma_2) \\ f(u_{r_4}, v_{r_4}, \beta_2)\sin(\gamma_2) \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Note that in this example we have two turbines and each turbine works on two cells.

Linearising around a point $\bar{x_k}$ results in

$$A(\bar{x_k})\delta x_{k+1} = \left(Q + \frac{\partial B_c(x_k)}{\partial x_k} - \frac{\partial A(x_k)}{\partial x_k}\bar{x}_{k+1} + \frac{\partial S_m}{\partial x_k}\right)\delta x_k + \frac{\partial S_m}{\partial \beta_k}\delta\beta_k + \frac{\partial S_m}{\partial \gamma_k}\delta\gamma_k$$

with deviation variables $\delta x_k, \delta\beta_k$ and $\delta\gamma_k$. Then we can define:

$$\frac{\partial B_c(x_k)}{\partial x_k} = B_c(x_k)\begin{pmatrix}\frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \cdots & \frac{\partial}{\partial x_N}\end{pmatrix} = \begin{pmatrix}\texttt{diag}\left(\frac{\partial a_{2,2}^{W_x}}{\partial u_{3,2}}, \frac{\partial a_{2,3}^{W_x}}{\partial u_{3,3}}, \ldots, \frac{\partial a_{2,N_y-1}^{W_x}}{\partial u_{3,2}}\right) & 0 \\ 0 & 0 \\ \texttt{diag}\left(\frac{\partial a_{1,3}^{W_y}}{\partial u_{3,2}}, \frac{\partial a_{1,4}^{W_y}}{\partial u_{3,3}}, \ldots, \frac{\partial a_{1,N_y-1}^{W_y}}{\partial u_{3,2}}\right) & 0 \\ 0 & 0\end{pmatrix}$$

and

$$\frac{\partial S_m(x_k)}{\partial x_k} = S_m(x_k)\begin{pmatrix}\frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \cdots & \frac{\partial}{\partial x_N}\end{pmatrix} = \begin{pmatrix}
0 & \ldots & & & & & \ldots & & & & & & & & & 0 & & \\
0 & \ldots & & & & 0 & \ldots & & & & & & & & & 0 & & \\
0 & -\frac{\partial f\cos(\gamma_1)}{\partial u_{r_1}} & 0 & \ldots & & 0 & \ldots & & 0 & -\frac{\partial f\cos(\gamma_1)}{\partial v_{r_1}} & \ldots & & 0 & & & & 0 & & \\
0 & \ldots & -\frac{\partial f\cos(\gamma_1)}{\partial u_{r_2}} & 0 & \ldots & 0 & \ldots & & & \ldots & -\frac{\partial f\cos(\gamma_1)}{\partial v_{r_2}} & & 0 & & & & & & \\
0 & \ldots & \ldots & & & 0 & \ldots & & & \ldots & \ldots & & 0 & & & & & & \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & \ldots & 0 & & & & & & & & & \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & \ldots & 0 & & & & & & & & & \\
0 & \ldots & 0 & 0 & \ldots & -\frac{\partial f\cos(\gamma_2)}{\partial u_{r_3}} & & & & & & & -\frac{\partial f\cos(\gamma_2)}{\partial v_{r_3}} & & & & & & \\
0 & \ldots & & 0 & \ldots & & -\frac{\partial f\cos(\gamma_2)}{\partial u_{r_4}} & \ldots & & 0 & & & & & -\frac{\partial f\cos(\gamma_2)}{\partial v_{r_4}} & & & & \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & \ldots & 0 & & & & & & 0 & \ldots & \ldots & 0 \\
0 & \ldots & & & & 0 & \ldots & & & & & & & & & & 0 & & \\
0 & \frac{\partial f\sin(\gamma_1)}{\partial u_{r_1}} & 0 & \ldots & & 0 & \ldots & & 0 & \frac{\partial f\sin(\gamma_1)}{\partial v_{r_1}} & \ldots & & 0 & & & & & & \\
0 & \ldots & \frac{\partial f\sin(\gamma_1)}{\partial u_{r_2}} & 0 & \ldots & 0 & \ldots & & & \ldots & \frac{\partial f\sin(\gamma_1)}{\partial v_{r_2}} & & 0 & & & & & & \\
0 & \ldots & \ldots & 0 & & 0 & \ldots & & & \ldots & \ldots & & 0 & & & & & & \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & \ldots & 0 & & & & & & & & & \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & \ldots & 0 & & & & & & & & & \\
0 & \ldots & 0 & 0 & \ldots & \frac{\partial f\sin(\gamma_2)}{\partial u_{r_3}} & & & & & & & \frac{\partial f\sin(\gamma_2)}{\partial v_{r_3}} & & & & & & \\
0 & \ldots & & 0 & \ldots & & \frac{\partial f\sin(\gamma_2)}{\partial u_{r_4}} & \ldots & & 0 & & & & & \frac{\partial f\sin(\gamma_2)}{\partial v_{r_4}} & & & & \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & \ldots & 0 & & & & & & & & &
\end{pmatrix}$$

with the derivatives as defined above. Furthermore we have

$$\frac{\partial S_m(\beta)}{\partial \beta} = S_m(\beta)\begin{pmatrix} \frac{\partial}{\partial \beta_1} & \frac{\partial}{\partial \beta_2} \end{pmatrix}$$

$$= \begin{pmatrix}
0 & 0 \\
\vdots & \vdots \\
0 & 0 \\
-\frac{\partial f(\beta_1)\cos(\gamma_1)}{\partial \beta_1} & 0 \\
-\frac{\partial f(\beta_1)\cos(\gamma_1)}{\partial \beta_1} & 0 \\
0 & 0 \\
0 & 0 \\
\vdots & \vdots \\
0 & 0 \\
0 & -\frac{\partial f(\beta_2)\cos(\gamma_2)}{\partial \beta_2} \\
0 & -\frac{\partial f(\beta_2)\cos(\gamma_2)}{\partial \beta_2} \\
0 & 0 \\
\vdots & \vdots \\
0 & 0 \\
\frac{\partial f(\beta_1)\sin(\gamma_1)}{\partial \beta_1} & 0 \\
\frac{\partial f(\beta_1)\sin(\gamma_1)}{\partial \beta_1} & 0 \\
0 & 0 \\
0 & 0 \\
\vdots & \vdots \\
0 & 0 \\
0 & \frac{\partial f(\beta_2)\sin(\gamma_2)}{\partial \beta_2} \\
0 & \frac{\partial f(\beta_2)\sin(\gamma_2)}{\partial \beta_2} \\
0 & 0 \\
\vdots & \vdots \\
0 & 0
\end{pmatrix}$$

and

$$\frac{\partial S_m(\gamma)}{\partial \gamma} = S_m(\gamma)\begin{pmatrix} \frac{\partial}{\partial \gamma_1} & \frac{\partial}{\partial \gamma_2} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ -\frac{\partial f(\gamma_1)\cos(\gamma_1)}{\partial \gamma_1} & 0 \\ -\frac{\partial f(\gamma_1)\cos(\gamma_1)}{\partial \gamma_1} & 0 \\ 0 & 0 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ 0 & -\frac{\partial f(\gamma_2)\cos(\gamma_2)}{\partial \gamma_2} \\ 0 & -\frac{\partial f(\gamma_2)\cos(\gamma_2)}{\partial \gamma_2} \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ \frac{\partial f(\gamma_1)\sin(\gamma_1)}{\partial \gamma_1} & 0 \\ \frac{\partial f(\gamma_1)\sin(\gamma_1)}{\partial \gamma_1} & 0 \\ 0 & 0 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ 0 & \frac{\partial f(\gamma_2)\sin(\gamma_2)}{\partial \gamma_2} \\ 0 & \frac{\partial f(\gamma_2)\sin(\gamma_2)}{\partial \gamma_2} \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \end{pmatrix}$$